



Supplementary Materials for

Cortical glutamatergic and GABAergic inputs support learning-driven hippocampal stability

Vincent Robert *et al.*

Corresponding author: Jayeeta Basu, jayeeta.basu@nyulangone.org

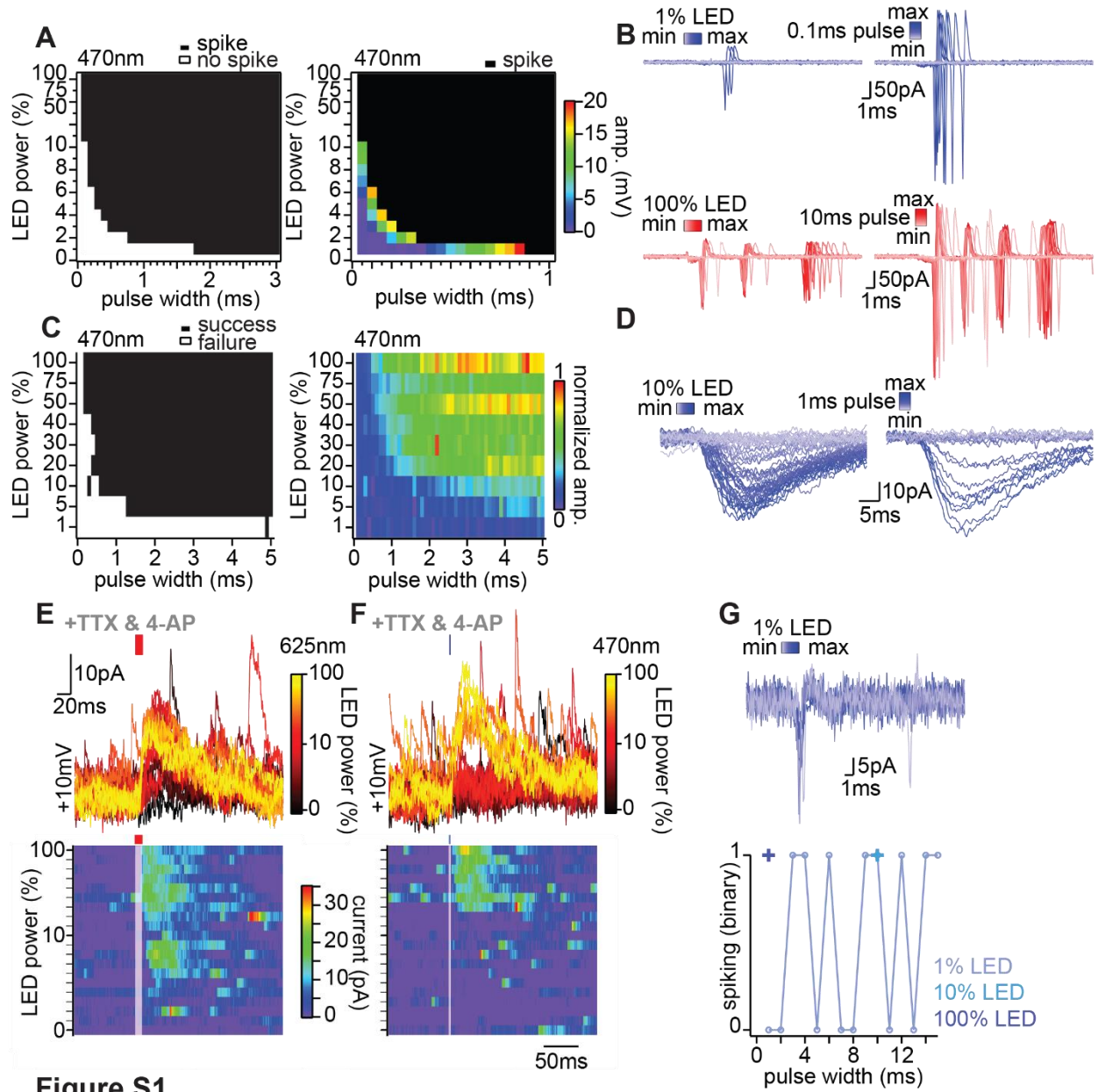
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The PDF file includes:

Figs. S1 to S19
Tables S1 to S4

Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist



Sample traces (top) and spiking quantification (bottom) of Chronos-expressing LEC_{GLU} neurons recorded in cell-attached configuration with increasing pulse width.

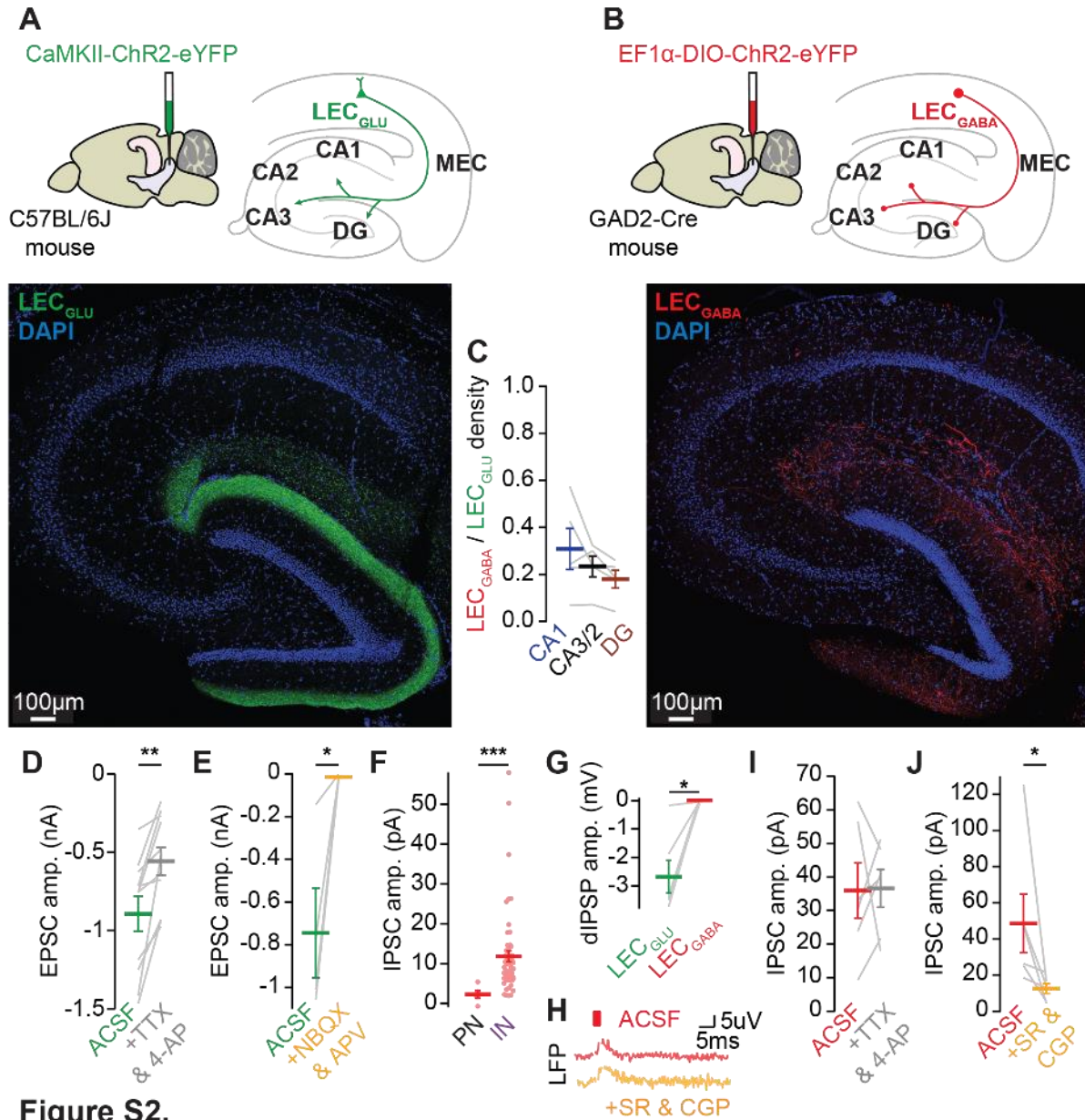


Figure S2.

Fig S2. Pharmacology of LEC_{GLU} and LEC_{GABA} synaptic transmission in CA3.

(A) Top, AAV injection of CaMKII-driven opsin into LEC of C57BL/6J mice allows expression of ChR2-GFP in LEC_{GLU}. Bottom, sample LEC_{GLU} projections (eYFP, green) to the hippocampus, with DAPI staining (blue). (B) Top, AAV injection of Syn-driven-flexed opsin into LEC of GAD2-Cre mice allows expression of ChR2-eYFP in LEC_{GABA}. Right, sample LEC_{GABA} projections (eYFP, red) to the hippocampus, with DAPI staining (blue). (C) LEC_{GABA} over LEC_{GLU} mean fluorescence density ratio across hippocampal subfields (0.31 ± 0.09 in CA1, 0.23 ± 0.04 in CA2/3, 0.18 ± 0.04 in DG, $n = 5$, repeated-measures ANOVA, $p = 0.080$). (D) LEC_{GLU}-evoked CA3 PN EPSC amplitude before and after application of TTX & 4-AP (-893.7 ± 111.8 pA in ACSF and -557.6 ± 89.5 pA in TTX & 4-AP, $n = 11$, paired-T test, $p = 0.001$). (E) LEC_{GLU}-evoked CA3 PN EPSC amplitude before and after application of NBQX & APV (-744.9 ± 210.0 pA in ACSF and -14.7 ± 5.5 pA in NBQX & APV, $n = 4$, paired-T test, $p = 0.038$). (F) LEC_{GABA}-evoked IPSC

amplitude in CA3 PNs and INs (11.8 ± 1.4 pA for INs, $n = 55$; 2.3 ± 1.0 pA for PNs, $n = 5$, Mann-Whitney U test, $p < 0.001$). **(G)** CA3 PN dendritic IPSP amplitude evoked by stimulation of LEC_{GLU} and LEC_{GABA} ($n = 6$, Wilcoxon signed-rank test, $p = 0.031$). **(H)** Sample traces of CA3 SLM LFP upon LEC_{GABA} stimulation in ACSF (red) and SR & CGP (orange), note that the positive voltage deflection seen in ACSF remain in SR & CGP therefore likely indicating a photoelectric artifact rather than a synaptic event. **(I)** LEC_{GABA} -evoked CA3 IN IPSC amplitude before and after application of TTX & 4-AP (35.9 ± 8.2 pA in ACSF and 36.6 ± 5.6 pA in TTX & 4-AP, $n = 6$, paired-T test, $p = 0.945$). **(J)** LEC_{GABA} -evoked CA3 IN IPSC amplitude before and after application of SR & CGP (48.6 ± 16.2 pA in ACSF and 12.5 ± 2.7 pA in SR & CGP, $n = 6$, Wilcoxon signed-rank test, $p = 0.031$). Error bars represent SEM.

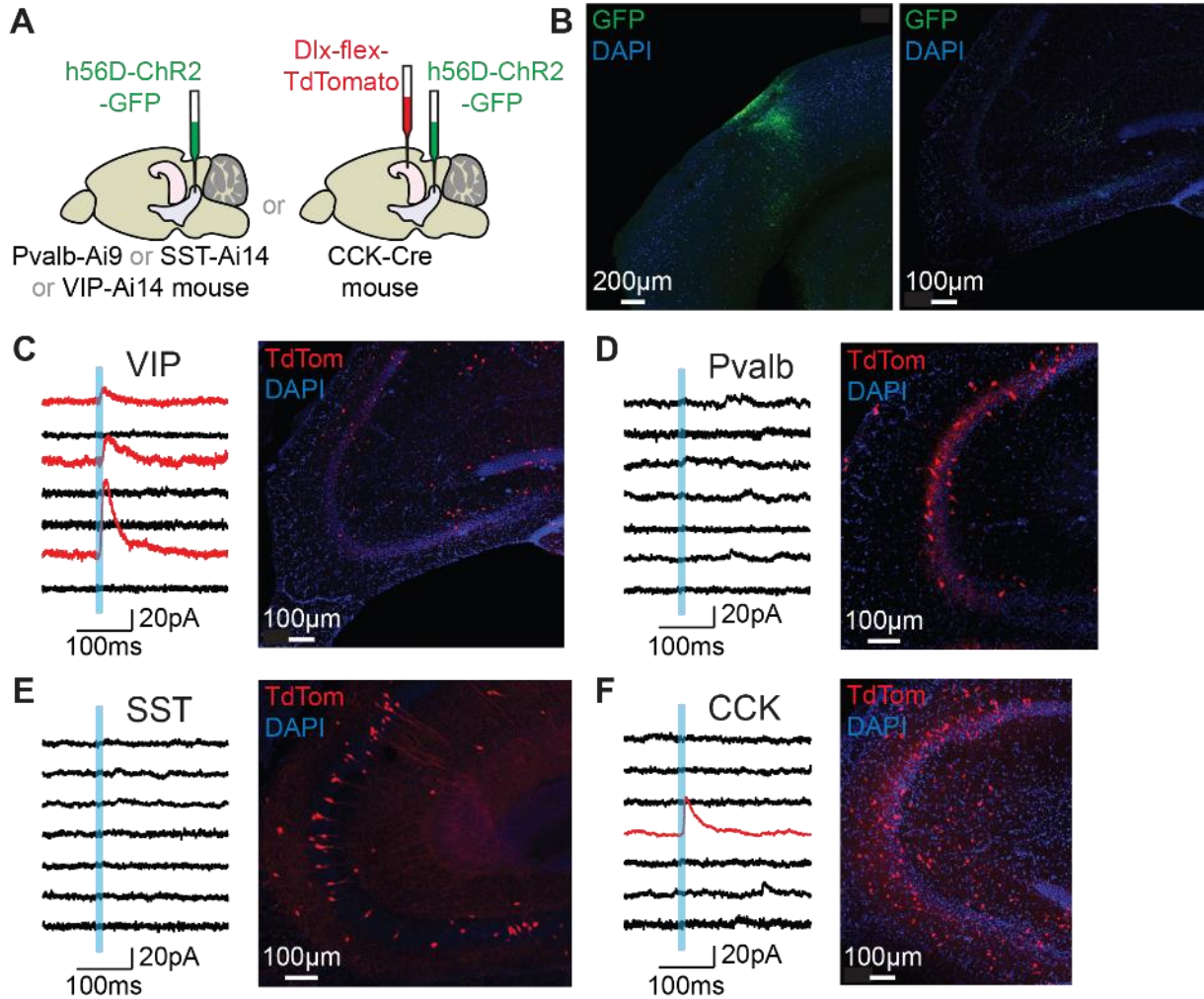


Figure S3.

Fig S3. Mapping LEC_{GABA} inputs to CA3 INs.

(A) AAV injection of h56D-driven opsin into LEC of Pvalb, SST or VIP reporter mice or CCK-Cre mice with CA3 injection of Dlx-driven reporter allows expression of ChR2-GFP in LEC_{GABA} and mapping of its IN subtypes CA3 targets with *ex vivo* patch-clamp electrophysiology. (B) Sample injection site of AAV targeting LEC_{GABA} (GFP, green, left) neurons and their projections to CA3 (right), with DAPI staining (blue). (C) Left, sample traces of seven CA3 VIP-expressing INs under voltage-clamp at +10mV upon LEC_{GABA} stimulation showing unresponsive (black) and responsive (red) post-synaptic cells (each trace is the average response of a single cell). Right, CA3 VIP-expressing INs (TdTomato, red) with DAPI staining (blue). (D) Same as (C) but with Pvalb-expressing INs. (E) Same as (C) but with SST-expressing INs. (F) Same as (C) but with CCK-expressing INs.

Text S1. *LEC_{GLU} can recruit DG and RC local inputs to yield CA3 PN output*

Multipeaked EPSCs evoked with optogenetic stimulation of LEC_{GLU} above 3 % (0.21 mW/mm²) LED intensity suggested that LEC_{GLU} may recruit polysynaptic excitation from DG and/or CA3/2. We confirmed this by performing extracellular field potential recordings in DG GC and CA3 SP layers which readily displayed population spikes as LEC_{GLU} stimulation intensity exceeded 5 % (0.33 mW/mm²) (Fig S4F-H), thus indicating that LEC_{GLU} can indeed recruit DG and RC as additional sources of excitation onto CA3 PNs. Conversely, these observations suggest that monosynaptic LEC_{GLU} input could be probed below 3 % (0.21 mW/mm²) LED power threshold in our experimental conditions (Fig 2B). Quantification of the monosynaptic and polysynaptic LEC_{GLU}, DG, and RC inputs onto CA3 PNs revealed substantial excitatory and inhibitory drives (Fig S4I-K). This led to significant compartment-specific depolarizations that remained subthreshold when inputs were stimulated in isolation with GABAergic transmission intact (Fig S4L-M). Altogether, these findings suggest that the dynamics of synaptic integration and relief from feedforward inhibition regulate CA3 PNs spike output in response to LEC_{GLU} input. Since no single input drove firing in CA3 PN (Fig 2A), but LEC_{GLU} may polysynaptically recruit local DG and RC inputs (Fig 2B, Fig S4F-H), we asked whether repeated or combined inputs could elicit CA3 PN spiking. First, we repeatedly stimulated LEC_{GLU} either in the monosynaptic (2 % light intensity, 0.14 mW/mm²) or polysynaptic (100 % light intensity, 3.76 mW/mm²) regime while recording PSPs and spikes from CA3 PNs soma or dendrites (Fig S4N-O). Monosynaptic activation of LEC_{GLU} yielded small depolarizations and virtually no spiking (Fig S4N-O), consistent with LEC_{GLU} targeting CA3 PNs distal dendrites and therefore generating PSPs that attenuate before reaching the soma. In contrast, polysynaptic activation of LEC_{GLU} induced PSP summation and spiking (Fig S4N-O), suggesting that recruitment of DG and/or RC inputs by LEC_{GLU} could drive CA3 dendritic spiking and somatic output. To confirm this, we probed whether polysynaptic (100 % light intensity, 3.76 mW/mm²) activation of LEC_{GLU} would occlude additive effects of pairing LEC_{GLU} with DG or RC. For this, we delivered repeated stimulations of LEC_{GLU}, DG, and RC either alone or combined with one another while monitoring PSPs and spikes in CA3 PN somas or dendrites. Although individual pathways elicited modest spiking in CA3 PNs, combining LEC_{GLU} with DG or RC did not yield significantly more firing than expected from the linear sum of each input alone (Fig S5A-D). These results reveal an overlap between polysynaptic LEC_{GLU} and DG or RC pathways, as expected if strong LEC_{GLU} stimulation recruits additional DG- or RC-driven excitation onto CA3 PN. Therefore, the ceiling effect of CA3 output in response to combined polysynaptic LEC_{GLU} and DG or RC suggests that LEC_{GLU} can physiologically recruit both DG and RC local inputs to drive activity in CA3 PNs.

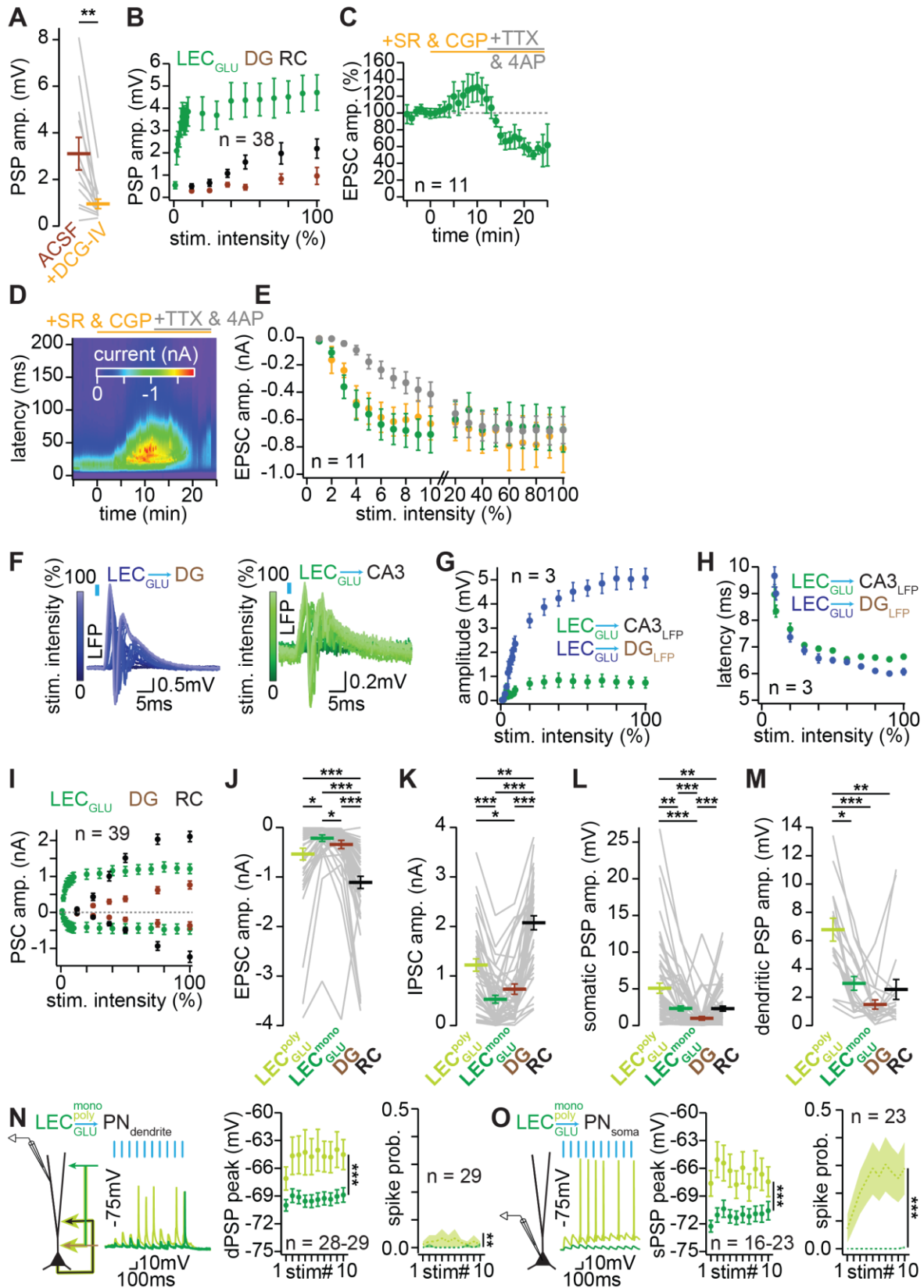


Figure S4.

Fig S4. LEC_{GLU} mono- and poly-synaptic transmission in CA3 (panels S4B, S4I, S4N & S4O replotted from 2A, 1C, 2E & 2F, respectively).

(A) DG-evoked CA3 PN PSP amplitude before and after application of DCG-IV (3.1 ± 0.7 mV in ACSF and 1.0 ± 0.2 mV in DCG-IV, $n = 12$, paired-T test, $p = 0.003$). (B) Input-output curves of LEC_{GLU}- (green, same data as Fig 2A), DG- (brown), and RC-evoked (black) CA3 PN PSP amplitudes. (C) Time-course of LEC_{GLU}-evoked CA3 PN EPSC amplitude. (D) Time x latency x current time-course of LEC_{GLU}-evoked CA3 PN EPSC. (E) Input-output curves of LEC_{GLU}-evoked CA3 PN EPSC amplitude in ACSF (green), SR & CGP (orange), and TTX & 4-AP (grey). (F) Sample traces of LEC_{GLU}-evoked LFP recordings in and DG GL (left, blue) CA3 SP (right, green). (G) Input-output curves of LEC_{GLU}-evoked population spike amplitude in CA3 SP (green) and DG GL (blue), ($n = 3$). (H) Input-output curves of LEC_{GLU}-evoked population spike latency in CA3 SP (green) and DG GL (blue) ($n = 3$). (I) Input-output curves of LEC_{GLU}- (green, same data as Fig 1C), DG- (brown), and RC-evoked (black) CA3 PN EPSC (negative) and IPSC (positive) amplitudes. (J) LEC_{GLU} polysynaptic- (light green), LEC_{GLU} monosynaptic- (dark green), DG- (brown), and RC-evoked (black) CA3 PN EPSC amplitudes (-536.3 ± 119.0 pA for LEC_{GLU} polysynaptic, -217.0 ± 65.5 pA for LEC_{GLU} monosynaptic, -343.4 ± 84.1 pA for DG, -1108.6 ± 123.5 pA for RC, $n = 50$, Friedman ANOVA, $p < 0.001$, Nemenyi *post hoc* test, LEC_{GLU} polysynaptic vs LEC_{GLU} monosynaptic, $p = 0.019$, LEC_{GLU} polysynaptic vs DG, $p = 0.990$, LEC_{GLU} polysynaptic vs RC, $p < 0.001$, LEC_{GLU} monosynaptic vs DG, $p = 0.044$, LEC_{GLU} monosynaptic vs RC, $p < 0.001$, DG vs RC, $p < 0.001$). (K) Same as (J) but with CA3 PN IPSC amplitudes (1221.1 ± 125.6 pA for LEC_{GLU} polysynaptic, 529.8 ± 79.6 pA for LEC_{GLU} monosynaptic, 733.2 ± 104.3 pA for DG, 2075.1 ± 141.1 pA for RC, $n = 41$, Friedman ANOVA, $p < 0.001$, Nemenyi *post hoc* test, LEC_{GLU} polysynaptic vs LEC_{GLU} monosynaptic, $p < 0.001$, LEC_{GLU} polysynaptic vs DG, $p = 0.045$, LEC_{GLU} polysynaptic vs RC, $p = 0.005$, LEC_{GLU} monosynaptic vs DG, $p = 0.591$, LEC_{GLU} monosynaptic vs RC, $p < 0.001$, DG vs RC, $p < 0.001$). (L) Same as (J) but with CA3 PN somatic PSP amplitudes (5.1 ± 0.7 mV for LEC_{GLU} polysynaptic, 2.3 ± 0.3 mV for LEC_{GLU} monosynaptic, 1.0 ± 0.2 mV for DG, 2.3 ± 0.3 mV for RC, $n = 61$, Friedman ANOVA, $p < 0.001$, Nemenyi *post hoc* test, LEC_{GLU} polysynaptic vs LEC_{GLU} monosynaptic, $p = 0.002$, LEC_{GLU} polysynaptic vs DG, $p < 0.001$, LEC_{GLU} polysynaptic vs RC, $p = 0.003$, LEC_{GLU} monosynaptic vs DG, $p < 0.001$, LEC_{GLU} monosynaptic vs RC, $p = 0.998$, DG vs RC, $p < 0.001$). (M) Same as (J) but with CA3 PN dendritic PSP amplitudes (6.8 ± 0.8 mV for LEC_{GLU} polysynaptic, 3.0 ± 0.5 mV for LEC_{GLU} monosynaptic, 1.5 ± 0.3 mV for DG, 2.6 ± 0.7 mV for RC, $n = 19$, Friedman ANOVA, $p < 0.001$, Nemenyi *post hoc* test, LEC_{GLU} polysynaptic vs LEC_{GLU} monosynaptic, $p = 0.030$, LEC_{GLU} polysynaptic vs DG, $p < 0.001$, LEC_{GLU} polysynaptic vs RC, $p = 0.001$, LEC_{GLU} monosynaptic vs DG, $p = 0.101$, LEC_{GLU} monosynaptic vs RC, $p = 0.685$, DG vs RC, $p = 0.617$). (N) Sample traces (left), PSP peak (middle) and spike probability (right) of CA3 PN dendritic responses to repeated stimulation of LEC_{GLU} monosynaptic (dark green) or polysynaptic (light green) input at 10 Hz (same LEC_{GLU} monosynaptic data as Fig 2E; PSP peak: $n = 28$ out of 29 samples because spikes were excluded from measurements, two-way ANOVA, polysynaptic vs monosynaptic, $p < 0.001$, pulse #, $p = 0.971$, polysynaptic vs monosynaptic x pulse #, $p = 0.999$; spike probability: $n = 29$, two-way ANOVA, polysynaptic vs monosynaptic, $p = 0.008$, pulse #, $p > 0.999$, polysynaptic vs monosynaptic x pulse #, $p = 0.991$). (O) Same as (N) but with CA3 PN somatic responses (same LEC_{GLU} monosynaptic data as Fig 2F; PSP peak: $n = 16$ out of 23 samples because spikes were excluded from measurements, two-way ANOVA, polysynaptic vs monosynaptic, $p < 0.001$, pulse #, $p = 0.825$, polysynaptic vs monosynaptic x pulse #, $p = 0.989$; spike probability: $n = 23$, two-

way ANOVA, polysynaptic vs monosynaptic, $p < 0.001$, pulse #, $p = 0.618$, polysynaptic vs monosynaptic x pulse #, $p = 0.633$). Error bars represent SEM.

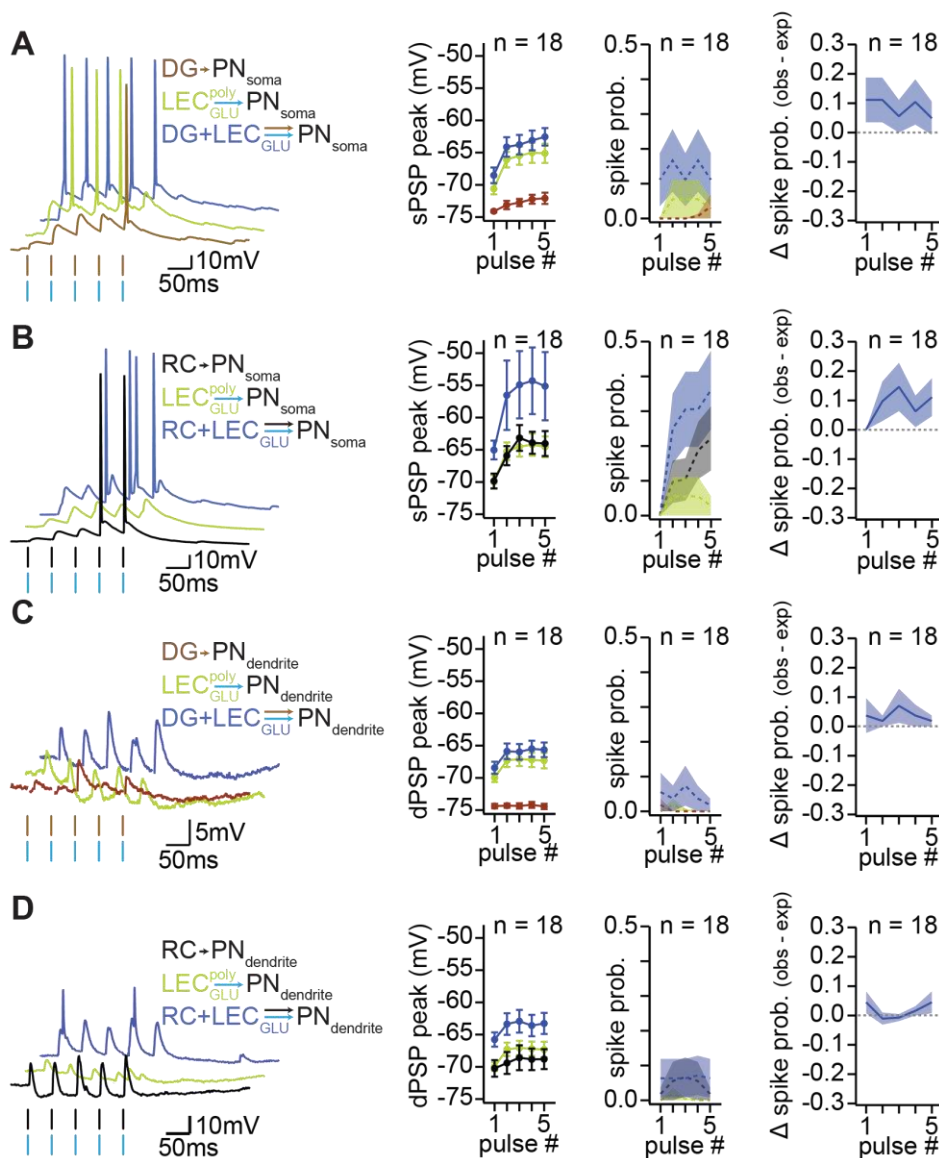


Figure S5.

Fig S5. LEC_{GLU} can recruit DG and RC inputs to yield CA3 PN output.

(A) Left, sample traces of CA3 PN somatic responses to repeated stimulations of LEC_{GLU} (green), DG (brown), and LEC_{GLU} + DG (blue) inputs at 20Hz. Middle, LEC_{GLU}- (green), DG- (brown), and LEC_{GLU} + DG-evoked (blue) CA3 PN somatic PSP peak (closed circles) and spike probability (dashed lines). Right, delta spike probability between observed DG + LEC_{GLU} stimulation versus expected from linear sum of these inputs (observed vs expected spike probability: n = 18, two-way ANOVA, observed vs expected, p = 0.050, pulse #, p = 0.915, observed vs expected x pulse #, p = 0.981). (B) Same as (A) but with RC stimulation (black) instead of DG (observed vs expected spike probability: n = 18, two-way ANOVA, observed vs expected, p = 0.146, pulse #, p = 0.008, observed vs expected x pulse #, p = 0.944). (C) Same as (A) but with CA3 PN dendritic responses (observed vs expected spike probability: n = 18, two-way ANOVA, observed vs expected, p = 0.074, pulse #, p = 0.871, observed vs expected x pulse #, p = 0.928). (D) Same as (A) but with

CA3 PN dendritic responses (observed vs expected spike probability: $n = 18$, two-way ANOVA, observed vs expected, $p = 0.599$, pulse #, $p = 0.967$, observed vs expected x pulse #, $p = 0.971$). Error bars represent SEM.

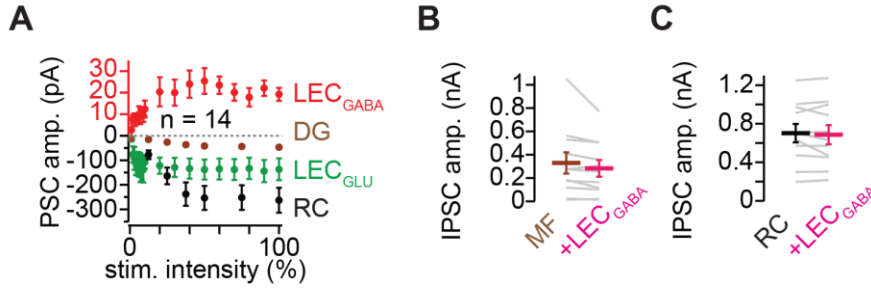


Figure S6.

Fig S6. LEC_{GABA} interaction with DG and RC inputs in CA3.

(A) Input-output curves of LEC_{GLU}- (green), LEC_{GABA}- (red), DG- (brown), and RC-evoked (black) CA3 PN EPSC (negative) and IPSC (positive) amplitudes (EPSC amplitude: -391.4 ± 243.0 pA for LEC_{GLU}, $n = 6$, -47.5 ± 8.0 for DG, $n = 50$, -329.8 ± 82.9 for RC, $n = 39$, Kruskal-Wallis ANOVA, $p < 0.001$, Dunn-Holland-Wolfe *post hoc* test, LEC_{GLU} vs DG, $p = 0.167$, LEC_{GLU} vs RC, $p = 0.973$, DG vs RC, $p = 0.012$). (B) CA3 PN IPSC amplitude evoked by DG with or without LEC_{GABA} stimulation (329.4 ± 90.9 pA for DG and 283.0 ± 71.5 pA for DG + LEC_{GABA}, $n = 11$, paired-T test, $p = 0.092$). (C) Same as (B) but with RC (702.2 ± 95.3 pA for RC and 686.3 ± 99.6 pA for RC + LEC_{GABA}, $n = 11$, paired-T test, $p = 0.464$). Error bars represent SEM.

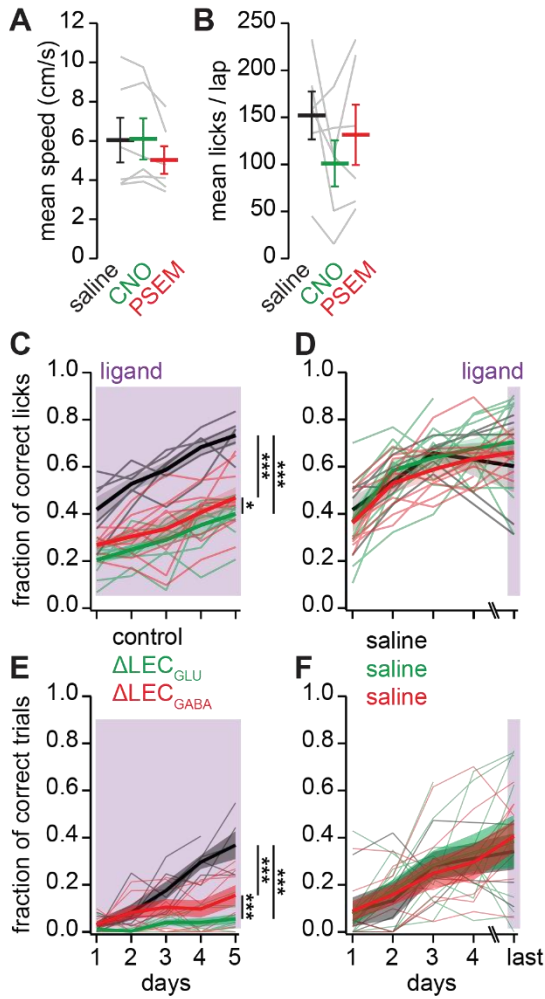


Figure S7.

Fig S7. Additional metrics related to head-fixed behavior experiments (panels S7C & S7D replotted from 4E & 4F, respectively).

(A) Mean speed of naïve animals injected with saline, CNO or PSEM during random foraging in a familiar environment (6.0 ± 1.1 cm/s in saline, 6.1 ± 1.1 cm/s in CNO, 5.0 ± 0.7 cm/s in PSEM, $n = 6$, repeated-measures ANOVA, $p = 0.067$). (B) Mean number of licks per lap of naïve animals injected with saline, CNO or PSEM during random foraging in a familiar environment (152 ± 25 licks/lap in saline, 101 ± 25 licks/lap in CNO, 131 ± 32 licks/lap in PSEM, $n = 6$, repeated-measures ANOVA, $p = 0.253$). (C) Fraction of licks in reward zones across days with ligand injections (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 10$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 9$, two-way ANOVA, treatment, $p < 0.001$, days, $p < 0.001$, treatment x days, $p = 0.795$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.014$; same data as Fig 4E). (D) Fraction of licks in reward zones across days with saline injections (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 10$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 9$, two-way ANOVA, treatment, $p = 0.873$, days, $p < 0.001$, treatment x days, $p = 0.996$; same data as Fig 4F). (E) Fraction of correct trials across days with ligand injections (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 10$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 9$, two-way ANOVA, treatment, $p < 0.001$, days, $p < 0.001$, treatment x days, $p < 0.001$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$). (F) Fraction of correct trials across days with saline injections (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 10$, $\Delta\text{LEC}_{\text{GABA}}$, $n =$

9, two-way ANOVA, treatment, $p = 0.623$, days, $p < 0.001$, treatment x days, $p = 0.855$). (**E** and **F**) Fraction of correct trials was computed as the fraction of laps in which mice had a fraction of licks restricted to the reward zone above their average licking rate per spatial bin during random foraging. Error bars represent SEM.

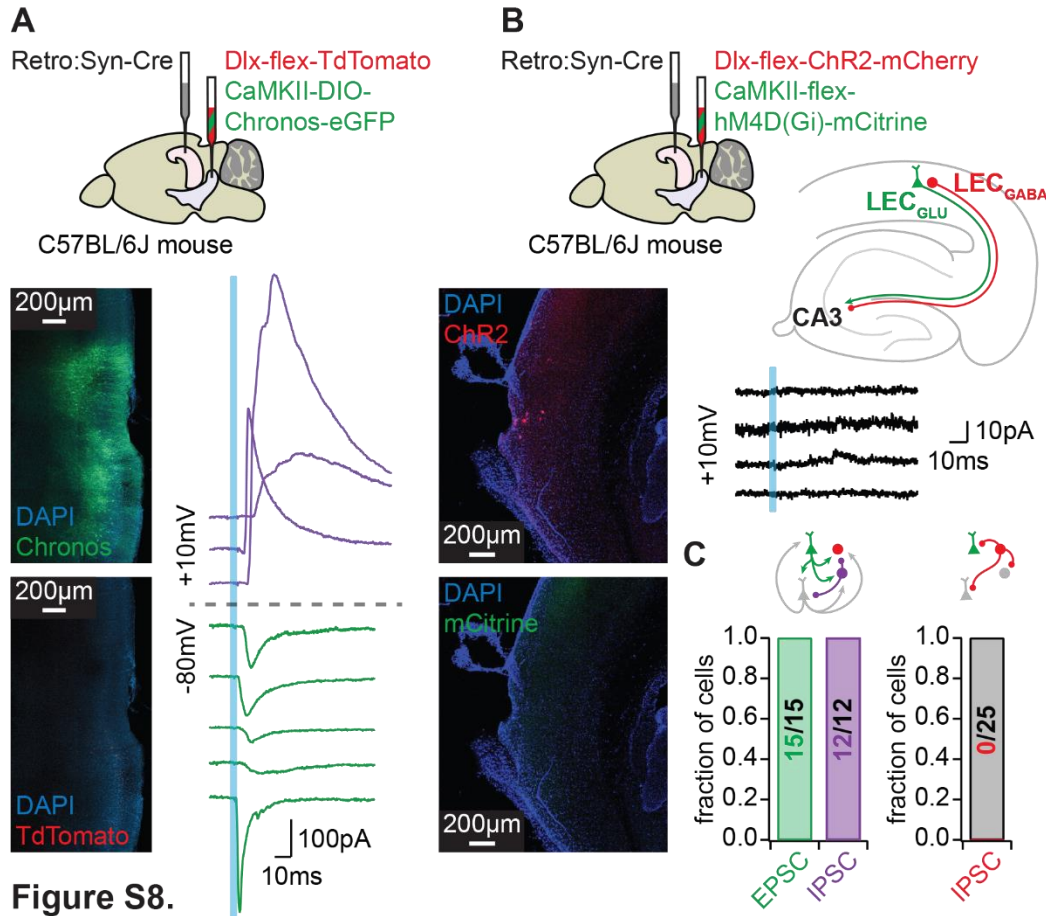


Figure S8.

Fig S8. CA3-projecting LEC_{GLU} and LEC_{GABA} local connectivity in LEC.

(A) Top, AAV injection of Syn-driven Cre-recombinase with retrograde tropism into CA3 and CaMKII-driven-flexed opsin and Dlx-driven-flexed reporter into LEC of C57BL/6J mice meant to allow expression of Chronos-eGFP in LEC_{GLU} and TdTomato in LEC_{GABA}, enabling mapping of local connectivity of CA3-projecting LEC_{GLU} in LEC with *ex vivo* patch-clamp electrophysiology. Bottom left, CA3-projecting retrogradely labelled LEC_{GLU} (eGFP, green) and LEC_{GABA} (TdTomato, red; note the inefficient expression with this viral strategy) neurons with DAPI staining (blue). Bottom right, sample traces of unlabeled LEC cells under voltage-clamp at +10mV (purple) or -80mV (green) upon stimulation of CA3-projecting LEC_{GLU} neurons (each trace is the average response of a single cell). (B) Top, AAV injection of Syn-driven Cre-recombinase with retrograde tropism into CA3 and Dlx-driven-flexed opsin and CaMKII-driven-flexed reporter into LEC of C57BL/6J mice meant to allow expression of ChR2-mCherry in LEC_{GABA} and mCitrine in LEC_{GLU}, enabling mapping of local connectivity of CA3-projecting LEC_{GABA} in LEC with *ex vivo* patch-clamp electrophysiology. Bottom left, CA3-projecting retrogradely labelled LEC_{GABA} (mCherry, red) and LEC_{GLU} (mCitrine, green; note the inefficient expression with this viral strategy) neurons with DAPI staining (blue). Bottom right, sample traces of unlabeled LEC cells under voltage-clamp at +10mV upon stimulation of CA3-projecting LEC_{GABA} neurons (each trace is the average response of a single cell). (C) Fraction of LEC cells connected by CA3-projecting LEC_{GLU} (left) or LEC_{GABA} (right).

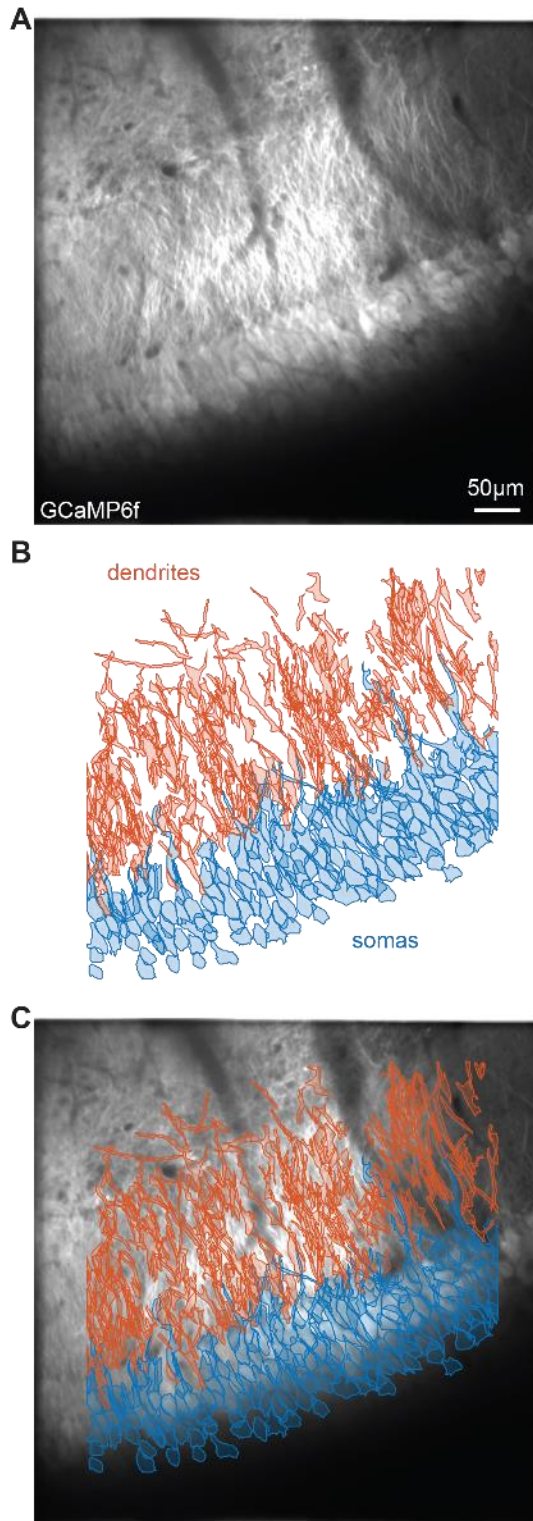


Figure S9.

Fig S9. Sample CA3 field of view and ROI footprints.

(A) Sample CA3 field of view. (B) Corresponding ROI footprints. (C) Overlay.

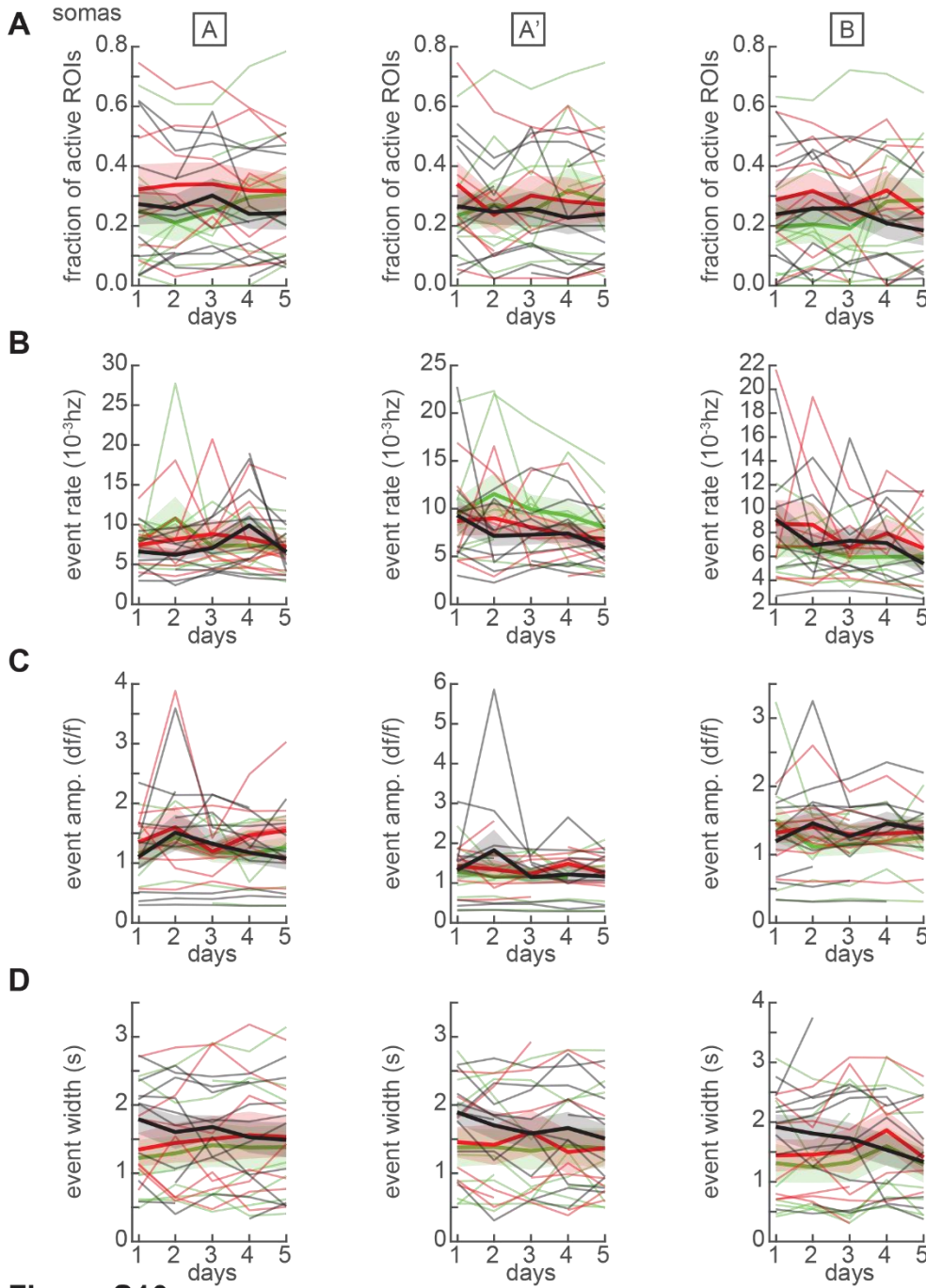


Figure S10.

Fig S10. Activity metrics of CA3 PN somas during *in vivo* two-photon calcium imaging.

(A) Fraction of active ROIs in the A (left), A' (middle) and B (right) environments in control (black, $n = 10$), Δ LECGLU (green, $n = 9$) and Δ LECGABA (red, $n = 8$) mice (A: two-way ANOVA, treatment, $p = 0.369$, days, $p = 0.987$, treatment $\times$ days, $p = 0.983$; A': two-way ANOVA, treatment, $p = 0.672$, days, $p = 0.988$, treatment $\times$ days, $p > 0.999$; B: two-way ANOVA, treatment, $p = 0.133$, days, $p = 0.999$, treatment $\times$ days, $p = 0.978$). (B) Same as (A) but with event rate (A: two-way ANOVA, treatment, $p = 0.587$, days, $p = 0.840$, treatment $\times$ days, $p = 0.837$; A': two-way ANOVA, treatment, $p = 0.112$, days, $p = 0.231$, treatment $\times$ days, $p = 0.740$; B: two-way

ANOVA, treatment, $p = 0.225$, days, $p = 0.296$, treatment x days, $p = 0.993$). (C) Same as (A) but with event amplitude (A: two-way ANOVA, treatment, $p = 0.523$, days, $p = 0.720$, treatment x days, $p = 0.984$; A': two-way ANOVA, treatment, $p = 0.723$, days, $p = 0.666$, treatment x days, $p = 0.480$; B: two-way ANOVA, treatment, $p = 0.302$, days, $p = 0.975$, treatment x days, $p = 0.976$). (D) Same as (A) but with event width (A: two-way ANOVA, treatment, $p = 0.423$, days, $p = 0.978$, treatment x days, $p = 0.977$; A': two-way ANOVA, treatment, $p = 0.374$, days, $p = 0.993$, treatment x days, $p = 0.982$; B: two-way ANOVA, treatment, $p = 0.597$, days, $p = 0.959$, treatment x days, $p = 0.966$). Error bars represent SEM.

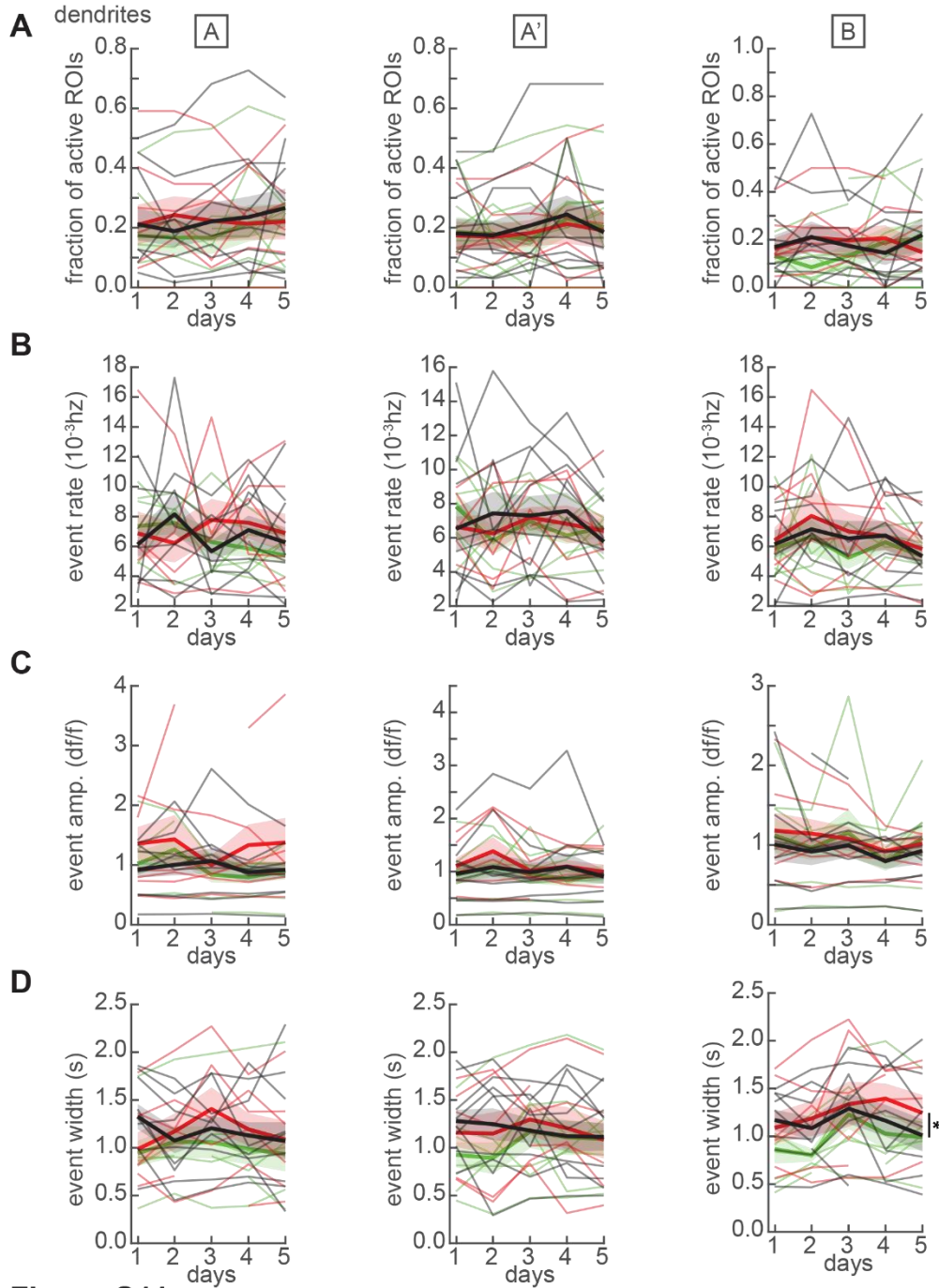


Figure S11.

Fig S11. Activity metrics of CA3 PN dendrites during *in vivo* two-photon calcium imaging.

(A) Fraction of active ROIs in the A (left), A' (middle) and B (right) environments in control (black, $n = 10$), ΔLECGLU (green, $n = 9$) and $\Delta\text{LECGABA}$ (red, $n = 8$) mice (A: two-way ANOVA, treatment, $p = 0.444$, days, $p = 0.976$, treatment $\times$ days, $p > 0.999$; A': two-way ANOVA, treatment, $p = 0.851$, days, $p = 0.988$, treatment $\times$ days, $p > 0.999$; B: two-way ANOVA, treatment, $p = 0.106$, days, $p = 0.751$, treatment $\times$ days, $p = 0.837$). (B) Same as (A) but with event rate (A: two-way ANOVA, treatment, $p = 0.404$, days, $p = 0.897$, treatment $\times$ days, $p = 0.902$; A': two-way ANOVA, treatment, $p = 0.616$, days, $p = 0.849$, treatment $\times$ days, $p = 0.963$; B: two-way

ANOVA, treatment, $p = 0.185$, days, $p = 0.688$, treatment x days, $p > 0.999$). (C) Same as (A) but with event amplitude (A: two-way ANOVA, treatment, $p = 0.300$, days, $p = 0.928$, treatment x days, $p = 0.628$; A': two-way ANOVA, treatment, $p = 0.477$, days, $p = 0.802$, treatment x days, $p = 0.983$; B: two-way ANOVA, treatment, $p = 0.980$, days, $p = 0.793$, treatment x days, $p = 0.970$). (D) Same as (A) but with event width (A: two-way ANOVA, treatment, $p = 0.203$, days, $p = 0.818$, treatment x days, $p = 0.916$; A': two-way ANOVA, treatment, $p = 0.141$, days, $p = 0.973$, treatment x days, $p = 0.989$; B: two-way ANOVA, treatment, $p = 0.026$, days, $p = 0.504$, treatment x days, $p = 0.999$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p = 0.328$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.322$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.019$). Error bars represent SEM.

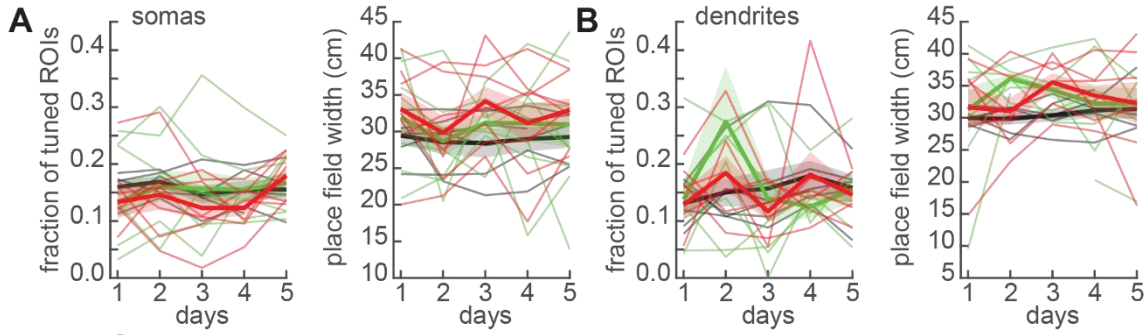


Figure S12.

Fig S12. Spatial tuning metrics of CA3 PN during *in vivo* two-photon calcium imaging.

(A) Fraction of tuned somas (left, control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p = 0.566$, days, $p = 0.697$, treatment $\times$ days, $p = 0.969$) and place field width (right, control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p = 0.129$, days, $p = 0.734$, treatment $\times$ days, $p = 0.997$). (B) Fraction of tuned dendrites (left, control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, PV: two-way ANOVA, treatment, $p = 0.855$, days, $p = 0.345$, treatment $\times$ days, $p = 0.699$) and place field width (right, control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, PV: two-way ANOVA, treatment, $p = 0.199$, days, $p = 0.825$, treatment $\times$ days, $p = 0.892$). Error bars represent SEM.

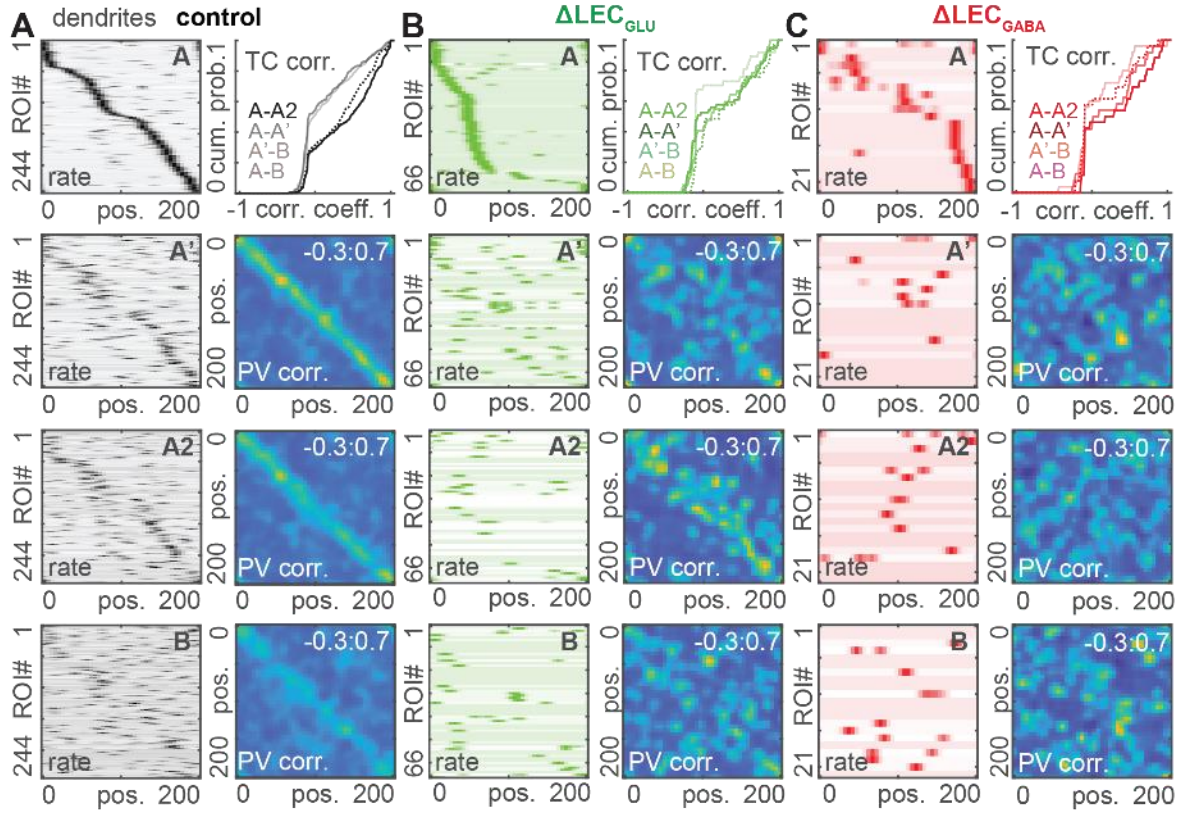


Figure S13.

Fig S13. Dendritic tuning during context-dependent CA3 remapping.

(A-C) Left, sample normalized rate maps of spatially tuned CA3 dendrites in the A (1st row), A' (2nd row), A2 (3rd row) and B (4th row) environments sorted according to their place field location in the A session. Right, sample spatial correlations reported as cumulative distribution of tuning curve (TC) correlation coefficients between A-A2 (dotted), A-A' (dark), A'-B (medium) and A-B (light) sessions (1st row), and population vector (PV) correlation matrix of A-A' (2nd row), A-A2 (3rd row) and A-B (4th row) sessions. (A) Control. (B) $\Delta\text{LEC}_{\text{GLU}}$. (C) $\Delta\text{LEC}_{\text{GABA}}$.

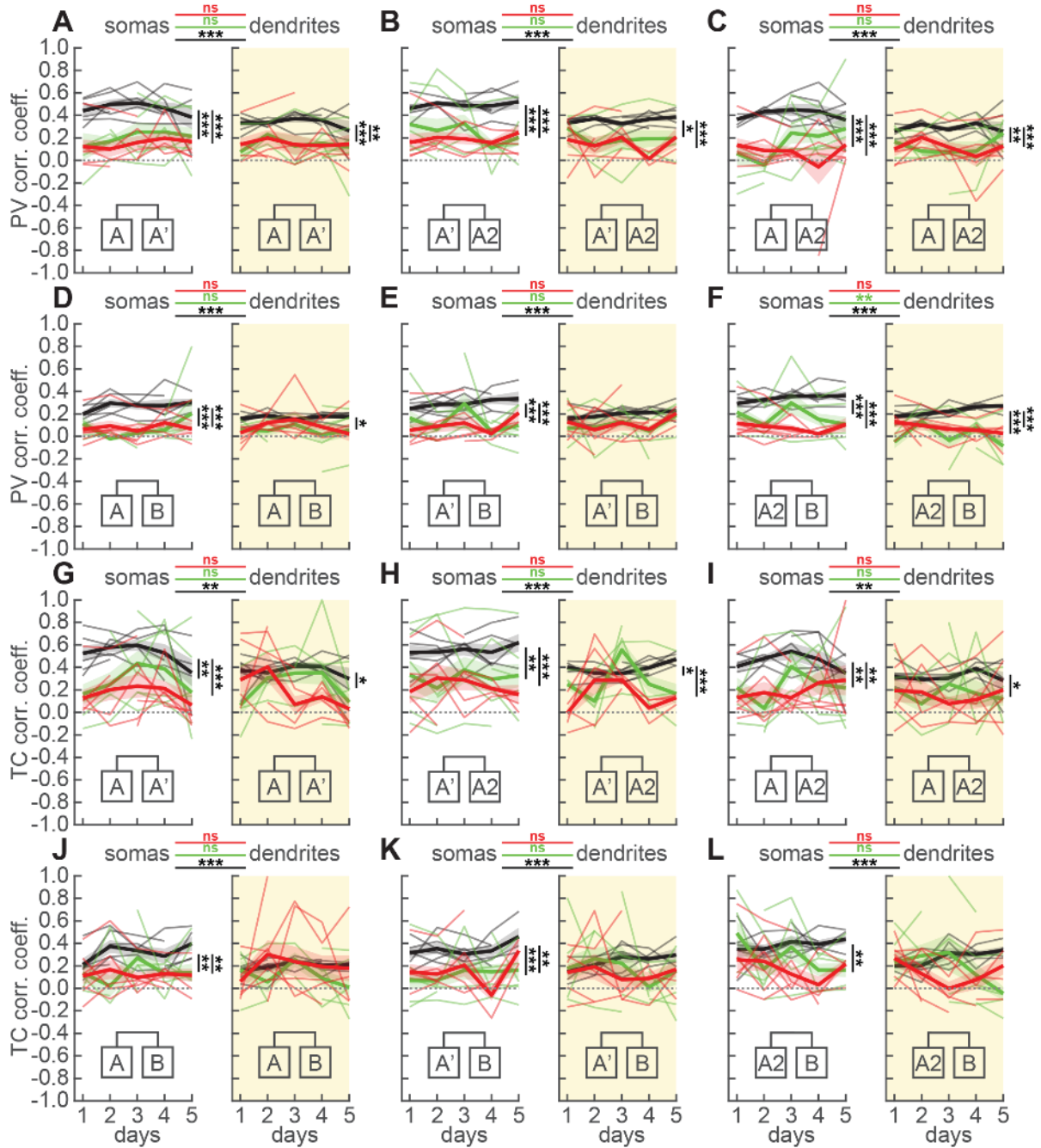


Figure S14.

Fig S14. Additional metrics related to context-dependent CA3 remapping (panels S14A-F replotted from 5F-K).

(A-F) Somatic (left) and dendritic (right) PV correlation coefficients across days (control, $n = 6$, ΔLECGLU , $n = 9$, $\Delta\text{LECGABA}$, $n = 8$, two-way ANOVAs, Tukey *post hoc* tests, Table 2, same data as Fig 5F-K). (A) A-A'. (B) A'-A2. (C) A-A2. (D) A-B. (E) A'-B. (F) A2-B. (G-L) Somatic (left) and dendritic (right) TC correlation coefficients across days (control, $n = 6$, ΔLECGLU , $n = 9$, $\Delta\text{LECGABA}$, $n = 8$, two-way ANOVAs, Tukey *post hoc* tests, Table S1). (G) A-A'. (H) A'-A2. (I) A-A2. (J) A-B. (K) A'-B. (L) A2-B. Error bars represent SEM.

Table S1. TC cross contexts comparisons p-values (two-way ANOVAs, Tukey *post hoc*).

somas	A-A'	A-A2	A-B	A'-A2	A'-B	A2-B
time	0.129	0.605	0.805	0.972	0.161	0.324
treatment	<0.001	0.001	<0.001	<0.001	<0.001	0.003
time x treatment	0.982	0.605	0.693	0.978	0.606	0.341
control vs $\Delta\text{LEC}_{\text{GLU}}$	0.006	0.007	0.005	0.006	<0.001	0.138
control vs $\Delta\text{LEC}_{\text{GABA}}$	<0.001	0.002	0.002	<0.001	0.002	0.002
$\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$	0.243	0.906	0.955	0.556	0.866	0.257
dendrites	A-A'	A-A2	A-B	A'-A2	A'-B	A2-B
time	0.208	0.975	0.778	0.046	0.815	0.901
treatment	0.029	0.032	0.475	<0.001	0.332	0.094
time x treatment	0.329	0.937	0.877	0.009	0.893	0.110
control vs $\Delta\text{LEC}_{\text{GLU}}$	0.141	0.074	0.507	0.034	0.564	0.366
control vs $\Delta\text{LEC}_{\text{GABA}}$	0.030	0.048	0.998	<0.001	0.317	0.081
$\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$	0.802	0.982	0.538	0.070	0.895	0.737
control	A-A'	A-A2	A-B	A'-A2	A'-B	A2-B
time	0.175	0.321	0.136	0.419	0.137	0.075
compartment	0.001	0.001	<0.001	<0.001	<0.001	<0.001
time x compartment	0.787	0.570	0.522	0.900	0.483	0.958
$\Delta\text{LEC}_{\text{GLU}}$	A-A'	A-A2	A-B	A'-A2	A'-B	A2-B
time	0.102	0.360	0.556	0.337	0.905	0.143
compartment	0.574	0.466	0.745	0.626	0.574	0.311
time x compartment	0.985	0.953	0.932	0.826	0.716	0.662
$\Delta\text{LEC}_{\text{GABA}}$	A-A'	A-A2	A-B	A'-A2	A'-B	A2-B
time	0.331	0.840	0.783	0.185	0.381	0.149
compartment	0.808	0.568	0.347	0.243	0.809	0.524
time x compartment	0.518	0.924	0.922	0.838	0.621	0.870

	somas			dendrites		
	control	$\Delta\text{LEC}_{\text{GLU}}$	$\Delta\text{LEC}_{\text{GABA}}$	control	$\Delta\text{LEC}_{\text{GLU}}$	$\Delta\text{LEC}_{\text{GABA}}$
time	0.350	0.100	0.796	0.020	0.073	0.307
context	<0.001	0.096	0.720	<0.001	0.630	0.946
time x context	0.544	0.990	0.827	0.449	0.896	0.817
A-A' vs A-A2	0.597	0.982	0.997	0.412	0.968	0.998
A-A' vs A-B	<0.001	0.505	0.987	<0.001	0.796	>0.999
A-A' vs A'-A2	0.920	0.999	0.942	0.939	0.999	0.996
A-A' vs A'-B	0.001	0.318	>0.999	<0.001	0.970	0.985
A-A' vs A2-B	0.024	>0.999	>0.999	0.005	0.993	0.984
A-A2 vs A-B	0.019	0.901	0.879	<0.001	0.993	0.994
A-A2 vs A'-A2	0.097	0.900	0.997	0.055	0.853	>0.999

A-A2 vs A'-B	0.177	0.777	0.982	0.052	>0.999	>0.999
A-A2 vs A2-B	0.663	0.995	0.999	0.531	>0.999	>0.999
A-B vs A'-A2	<0.001	0.291	0.642	<0.001	0.589	0.990
A-B vs A'-B	0.970	>0.999	0.999	0.681	0.993	0.968
A-B vs A2-B	0.553	0.633	0.979	0.083	0.976	0.966
A'-A2 vs A'-B	<0.001	0.151	0.867	<0.001	0.857	>0.999
A'-A2 vs A2-B	<0.001	0.997	0.968	<0.001	0.936	>0.999
A'-B vs A2-B	0.956	0.449	0.999	0.859	>0.999	>0.999

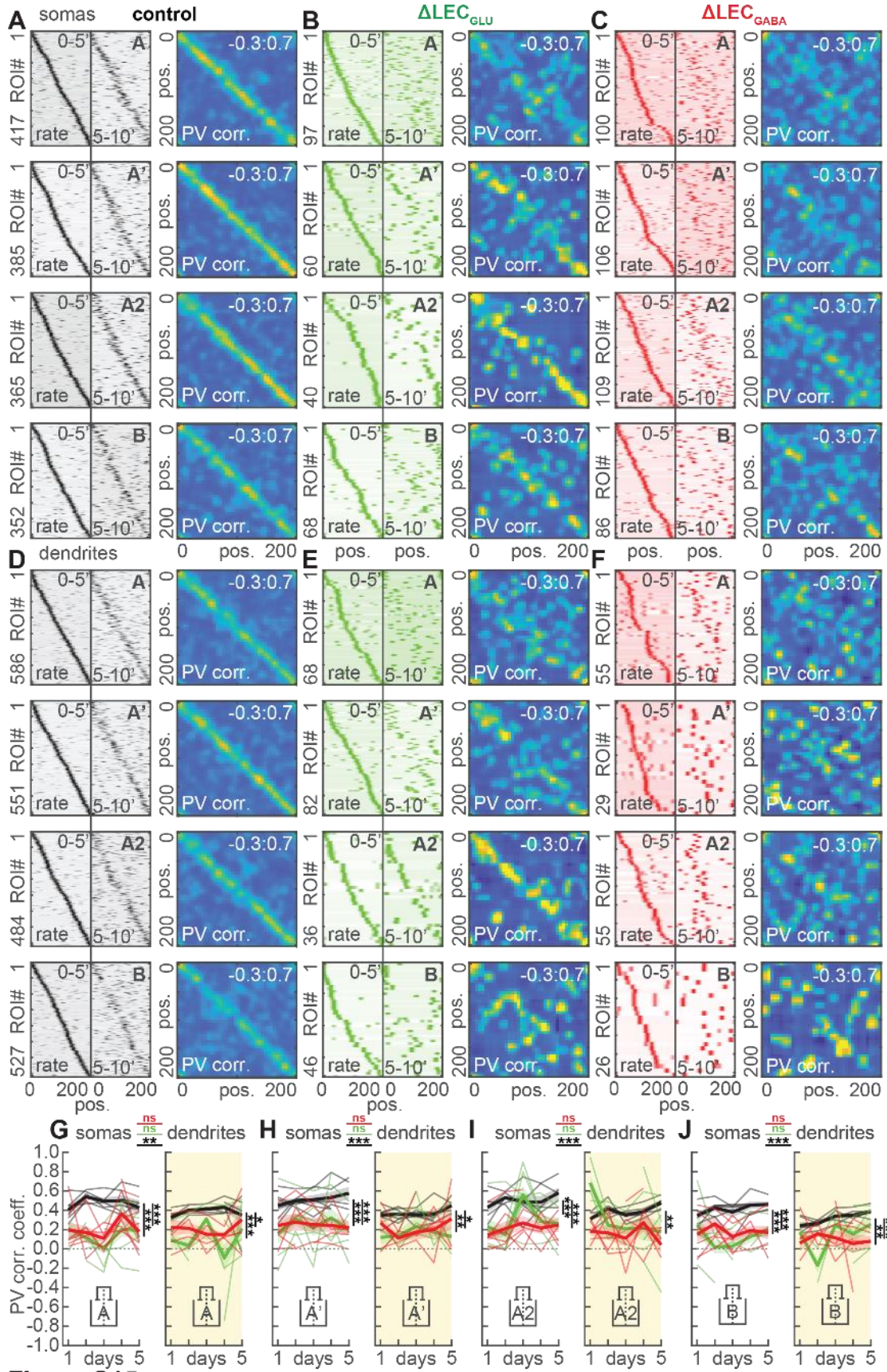


Fig S15. Within-session stability of CA3 spatial coding.

(A-C) Left, sample normalized rate maps of CA3 somas in the 1st and 2nd halves of sessions in the A (1st row), A' (2nd row), A2 (3rd row) and B (4th row) environments sorted according to their peak spatial activity in the 1st half of each session. Right, sample spatial correlations between the 1st and 2nd halves of each session reported as population vector (PV) correlation matrix. (A) Control. (B) $\Delta\text{LEC}_{\text{GLU}}$. (C) $\Delta\text{LEC}_{\text{GABA}}$. (D-F) Same as (A-C) but with CA3 dendrites. (G-J) Somatic (left) and dendritic (right) PV correlation coefficients across days (control, n = 6, $\Delta\text{LEC}_{\text{GLU}}$, n = 9, $\Delta\text{LEC}_{\text{GABA}}$, n = 8, two-way ANOVAs, Tukey *post hoc* tests, Table S2). (G) $A_{1^{\text{st half}}}^{\text{st half}}-A_{2^{\text{nd half}}}^{\text{nd half}}$. (H) $A'_{1^{\text{st half}}}^{\text{st half}}-A'_{2^{\text{nd half}}}^{\text{nd half}}$. (I) $A2_{1^{\text{st half}}}^{\text{st half}}-A2_{2^{\text{nd half}}}^{\text{nd half}}$. (J) $B_{1^{\text{st half}}}^{\text{st half}}-B_{2^{\text{nd half}}}^{\text{nd half}}$. Error bars represent SEM.

Table S2. PV cross halves comparisons p-values (two-way ANOVAs, Tukey *post hoc*).

somas	A ₁ st -A ₂ nd	A' ₁ st -A' ₂ nd	A2 ₁ st -A2 ₂ nd	B ₁ st -B ₂ nd
time	0.238	0.815	0.132	0.608
treatment	<0.001	<0.001	<0.001	<0.001
time x treatment	0.603	0.986	0.310	0.473
control vs ΔLEC _{GLU}	<0.001	<0.001	<0.001	<0.001
control vs ΔLEC _{GABA}	<0.001	<0.001	<0.001	<0.001
ΔLEC _{GLU} vs ΔLEC _{GABA}	0.498	0.997	0.306	0.506

dendrites	A ₁ st -A ₂ nd	A' ₁ st -A' ₂ nd	A2 ₁ st -A2 ₂ nd	B ₁ st -B ₂ nd
time	0.452	0.852	0.363	0.221
treatment	<0.001	0.001	0.002	<0.001
time x treatment	0.358	0.915	0.141	0.376
control vs ΔLEC _{GLU}	<0.001	0.001	0.460	0.006
control vs ΔLEC _{GABA}	0.017	0.022	0.002	<0.001
ΔLEC _{GLU} vs ΔLEC _{GABA}	0.288	0.629	0.076	0.791

control	A ₁ st -A ₂ nd	A' ₁ st -A' ₂ nd	A2 ₁ st -A2 ₂ nd	B ₁ st -B ₂ nd
time	0.115	0.297	0.023	0.011
compartment	0.007	<0.001	<0.001	<0.001
time x compartment	0.962	0.929	0.993	0.528

ΔLEC _{GLU}	A ₁ st -A ₂ nd	A' ₁ st -A' ₂ nd	A2 ₁ st -A2 ₂ nd	B ₁ st -B ₂ nd
time	0.669	0.835	0.254	0.134
compartment	0.632	0.236	0.911	0.923
time x compartment	0.136	0.922	0.102	0.558

ΔLEC _{GABA}	A ₁ st -A ₂ nd	A' ₁ st -A' ₂ nd	A2 ₁ st -A2 ₂ nd	B ₁ st -B ₂ nd
time	0.704	0.965	0.893	0.817
compartment	0.935	0.639	0.311	0.147
time x compartment	0.432	0.824	0.569	0.969

	somas			dendrites		
	control	ΔLEC _{GLU}	ΔLEC _{GABA}	control	ΔLEC _{GLU}	ΔLEC _{GABA}
time	0.051	0.805	0.800	0.003	0.252	0.969
context	0.018	0.034	0.710	0.009	0.115	0.272
time x context	0.933	0.189	0.920	0.224	0.392	0.887
A ₁ st -A ₂ nd vs A' ₁ st -A' ₂ nd	0.631	0.439	0.869	0.967	0.937	>0.999
A ₁ st -A ₂ nd vs A2 ₁ st -A2 ₂ nd	0.828	0.103	0.995	0.999	0.109	0.859
A ₁ st -A ₂ nd vs B ₁ st -B ₂ nd	0.284	0.989	0.972	0.024	0.996	0.362
A' ₁ st -A' ₂ nd vs A2 ₁ st -A2 ₂ nd	0.986	0.815	0.957	0.926	0.305	0.811
A' ₁ st -A' ₂ nd vs B ₁ st -B ₂ nd	0.019	0.287	0.660	0.078	0.992	0.301
A2 ₁ st -A2 ₂ nd vs B ₁ st -B ₂ nd	0.047	0.058	0.915	0.015	0.284	0.793

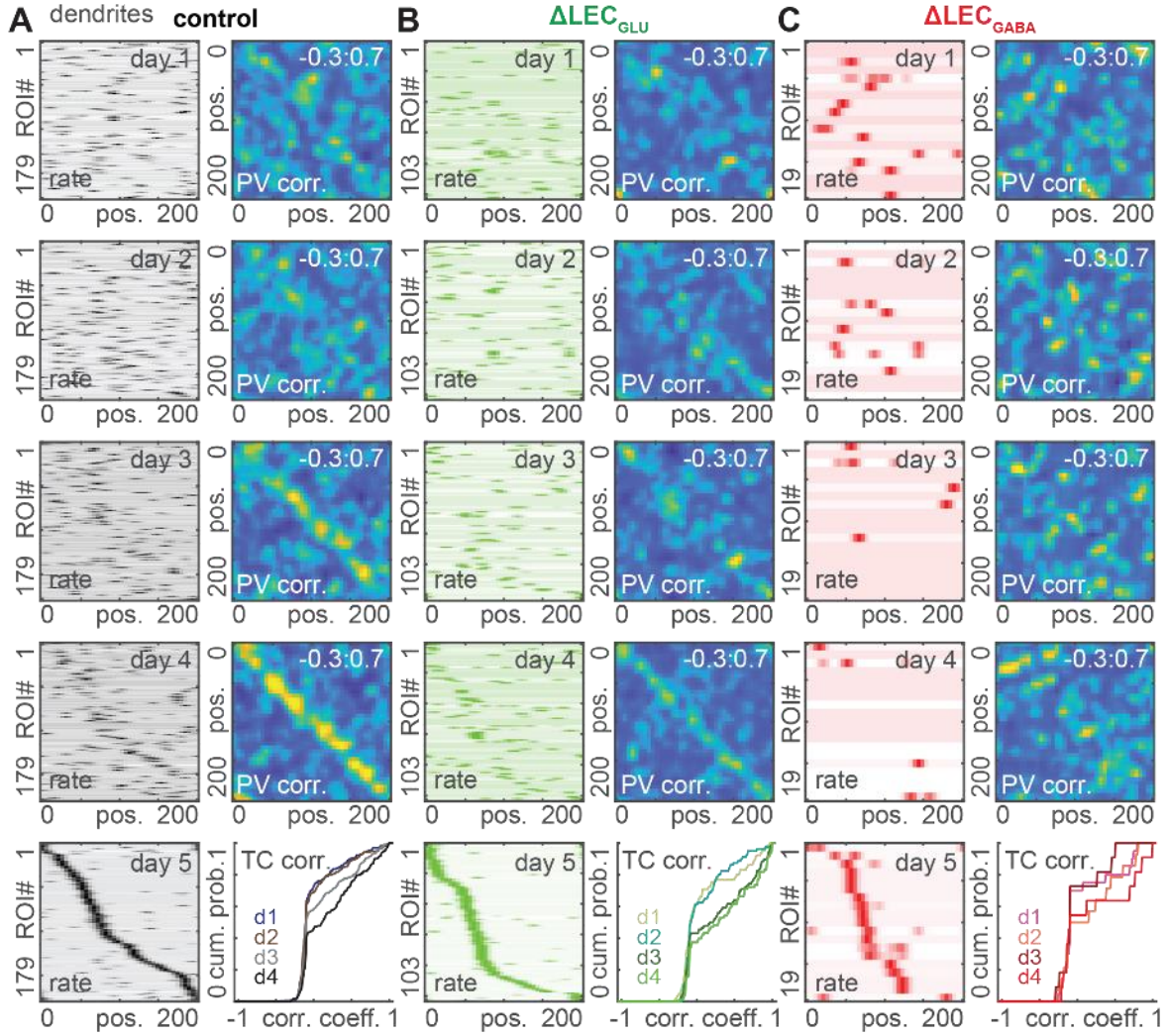


Figure S16.

Fig S16. CA3 dendritic tuning during learning.

(A-C) Left, sample normalized rate maps of spatially tuned CA3 dendrites in the A' environment across days 1-5 sorted according to their place field location from day 5. Right, sample spatial correlations across days 1-4 against day 5 reported as PV correlation matrix and cumulative distribution of TC correlation coefficients. (A) Control. (B) $\Delta\text{LEC}_{\text{GLU}}$. (C) $\Delta\text{LEC}_{\text{GABA}}$.

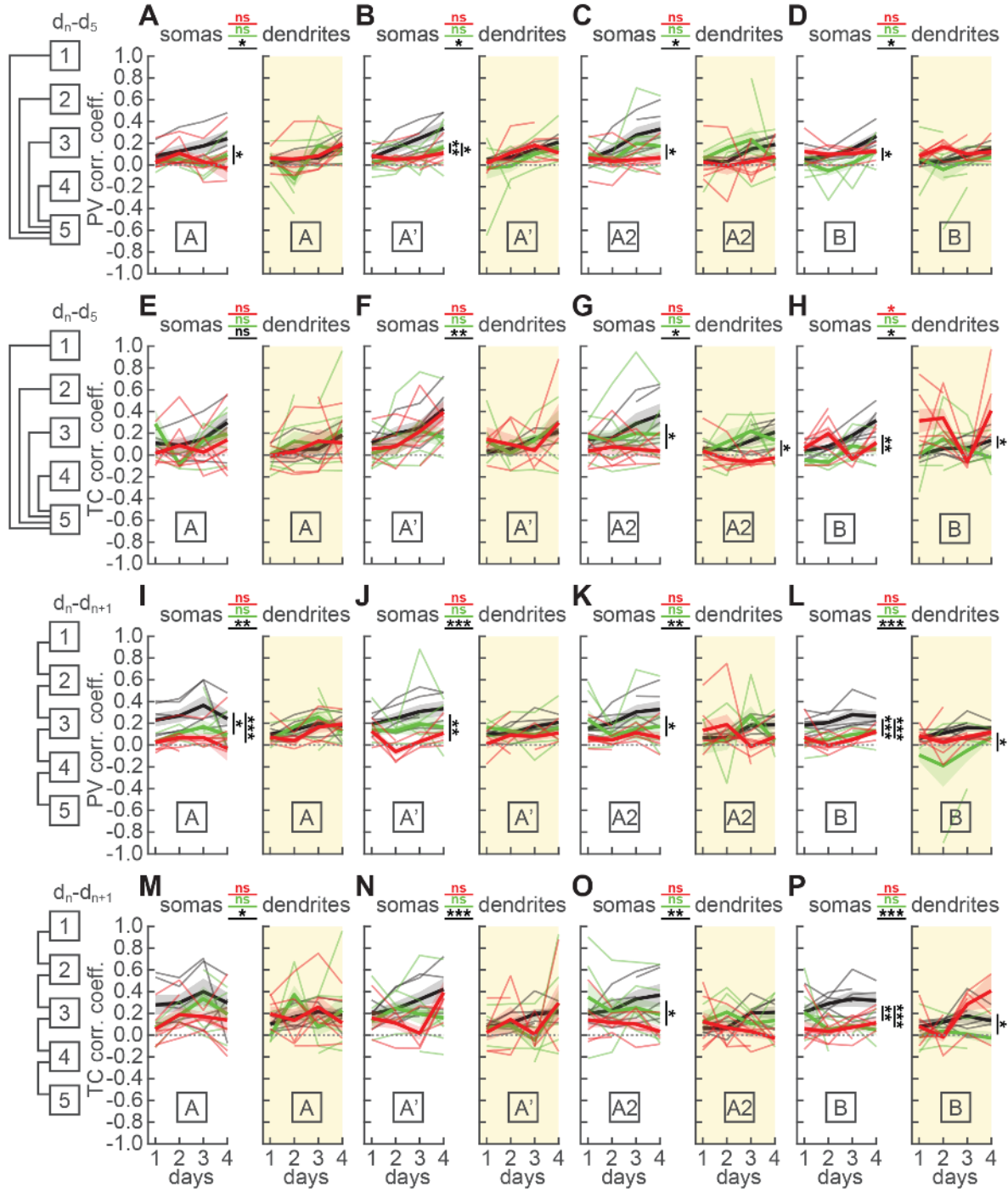


Figure S17.

Fig S17. Additional metrics related to learning-associated CA3 spatial coding (panels S17A-D & S17I-L replotted from 6D-G & 6H-K, respectively).

(A-D) Somatic (left) and dendritic (right) PV correlation coefficients across days relative to day 5 (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVAs, Tukey *post hoc* tests, Table 3, same data as Fig 6D-G). (A) A. (B) A'. (C) A2. (D) B. (E-H) Somatic (left) and dendritic (right) TC correlation coefficients across days relative to day 5 (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$,

$\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVAs, Tukey *post hoc* tests, Table S3). (E) A. (F) A'. (G) A2. (H) B. (I-L) Somatic (left) and dendritic (right) PV correlation coefficients across days between pairs of days (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVAs, Tukey *post hoc* tests, Table 4, same data as Fig 6H-K). (I) A. (J) A'. (K) A2. (L) B. (M-P) Somatic (left) and dendritic (right) TC correlation coefficients across days between pairs of days (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVAs, Tukey *post hoc* tests, Table S4). (M) A. (N) A'. (O) A2. (P) B. Error bars represent SEM.

Table S3. TC cross days to last comparisons p-values (two-way ANOVAs, Tukey *post hoc*).

somas	A	A'	A2	B
time	0.179	0.010	0.686	0.024
treatment	0.363	0.212	0.034	0.002
time x treatment	0.446	0.826	0.901	0.013
control vs $\Delta\text{LEC}_{\text{GLU}}$	0.919	0.197	0.626	0.002
control vs $\Delta\text{LEC}_{\text{GABA}}$	0.363	0.732	0.030	0.229
$\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$	0.592	0.565	0.184	0.108

dendrites	A	A'	A2	B
time	0.234	0.176	0.664	0.265
treatment	0.788	0.909	0.026	0.022
time x treatment	0.990	0.963	0.627	0.246
control vs $\Delta\text{LEC}_{\text{GLU}}$	0.844	0.922	0.979	0.958
control vs $\Delta\text{LEC}_{\text{GABA}}$	0.998	0.923	0.065	0.070
$\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$	0.805	>0.999	0.038	0.028

control	A	A'	A2	B
time	0.024	0.006	0.035	0.006
compartment	0.061	0.009	0.019	0.030
time x compartment	0.949	0.780	0.934	0.392

$\Delta\text{LEC}_{\text{GLU}}$	A	A'	A2	B
time	0.498	0.209	0.829	0.666
compartment	0.826	0.725	0.590	0.488
time x compartment	0.261	0.589	0.925	0.269

$\Delta\text{LEC}_{\text{GABA}}$	A	A'	A2	B
time	0.689	0.253	0.919	0.037
compartment	0.997	0.672	0.114	0.045
time x compartment	0.892	0.740	0.726	0.538

	somas			dendrites		
	control	$\Delta\text{LEC}_{\text{GLU}}$	$\Delta\text{LEC}_{\text{GABA}}$	control	$\Delta\text{LEC}_{\text{GLU}}$	$\Delta\text{LEC}_{\text{GABA}}$
time	<0.001	0.156	0.203	<0.001	0.549	0.213
context	0.242	0.106	0.152	0.295	0.662	0.017
time x context	0.998	0.693	0.530	0.976	0.836	0.579
A vs A'	0.484	0.997	0.223	0.595	0.983	0.800
A vs A2	0.494	0.943	0.993	0.528	0.999	0.714
A vs B	0.999	0.294	0.992	0.998	0.839	0.147
A' vs A2	>0.999	0.842	0.158	>0.999	0.996	0.231
A' vs B	0.434	0.327	0.431	0.515	0.634	0.648
A2 vs B	0.443	0.085	0.954	0.451	0.774	0.012

Table S4. TC cross days pairs comparisons p-values (two-way ANOVAs, Tukey *post hoc*).

somas	A	A'	A2	B
time	0.313	0.162	0.940	0.910
treatment	0.060	0.130	0.044	<0.001
time x treatment	0.980	0.302	0.788	0.558
control vs $\Delta\text{LEC}_{\text{GLU}}$	0.294	0.190	0.910	0.003
control vs $\Delta\text{LEC}_{\text{GABA}}$	0.050	0.183	0.050	<0.001
$\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$	0.744	0.978	0.114	0.857

dendrites	A	A'	A2	B
time	0.551	0.080	0.997	0.164
treatment	0.990	0.814	0.203	0.020
time x treatment	0.434	0.785	0.460	0.136
control vs $\Delta\text{LEC}_{\text{GLU}}$	>0.999	0.801	0.988	0.129
control vs $\Delta\text{LEC}_{\text{GABA}}$	0.991	0.931	0.292	0.541
$\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$	0.992	0.977	0.253	0.016

control	A	A'	A2	B
time	0.526	0.017	0.036	0.212
compartment	0.012	<0.001	0.003	<0.001
time x compartment	0.963	0.895	0.992	0.980

$\Delta\text{LEC}_{\text{GLU}}$	A	A'	A2	B
time	0.396	0.410	0.944	0.807
compartment	0.770	0.285	0.268	0.168
time x compartment	0.417	0.263	0.846	0.715

$\Delta\text{LEC}_{\text{GABA}}$	A	A'	A2	B
time	0.827	0.172	0.443	0.044
compartment	0.623	0.632	0.390	0.077
time x compartment	0.780	0.937	0.984	0.219

	somas			dendrites		
	control	$\Delta\text{LEC}_{\text{GLU}}$	$\Delta\text{LEC}_{\text{GABA}}$	control	$\Delta\text{LEC}_{\text{GLU}}$	$\Delta\text{LEC}_{\text{GABA}}$
time	0.061	0.864	0.621	<0.001	0.236	0.577
context	0.929	0.397	0.445	0.584	0.294	0.321
time x context	0.978	0.879	0.412	0.932	0.741	0.323
A vs A'	0.982	0.999	0.962	0.928	0.866	0.920
A vs A2	0.922	0.916	0.916	0.745	0.994	0.407
A vs B	0.954	0.722	0.666	0.550	0.280	0.999
A' vs A2	0.994	0.823	0.722	0.978	0.955	0.871
A' vs B	0.999	0.759	0.441	0.889	0.657	0.873
A2 vs B	0.999	0.318	0.971	0.989	0.395	0.350

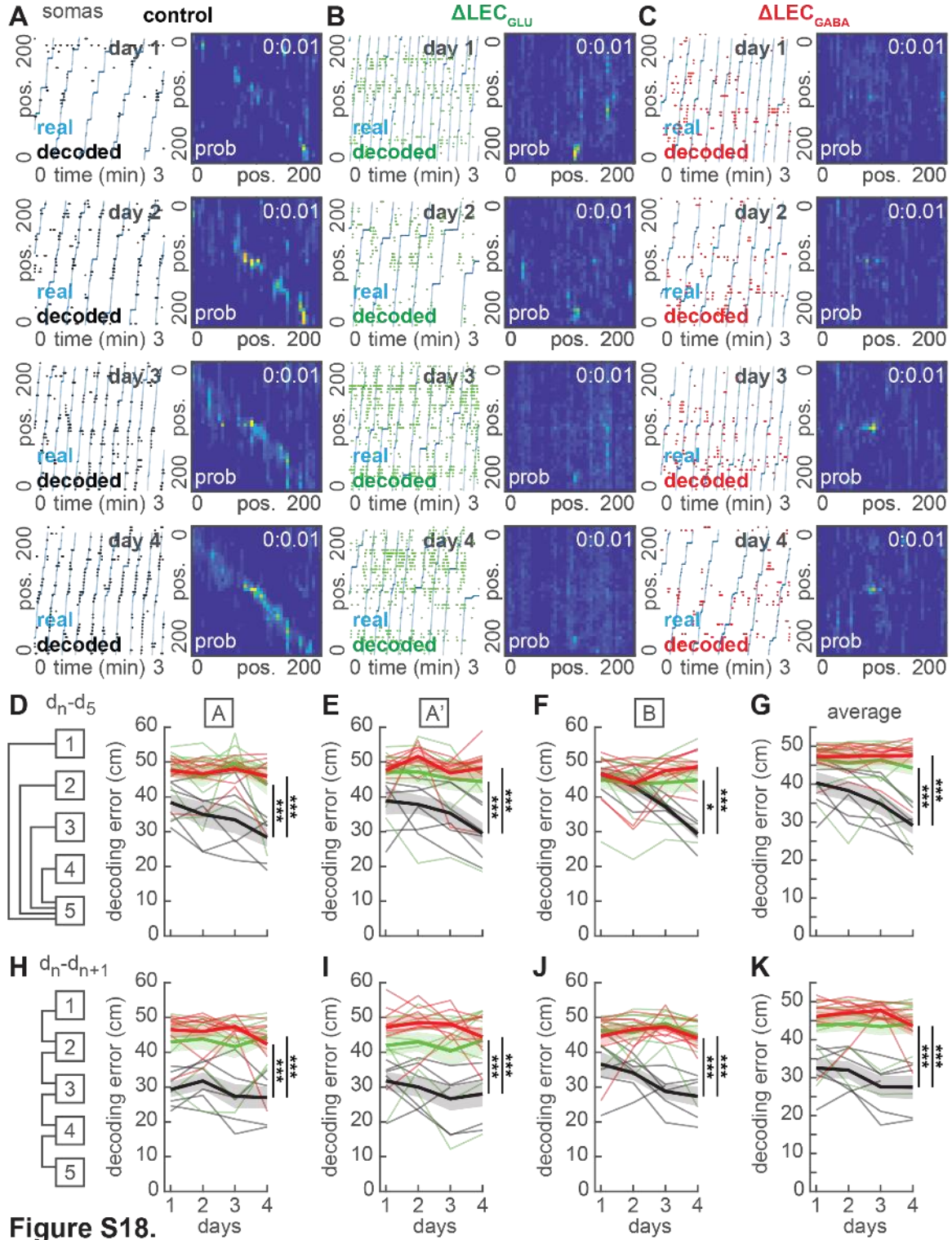


Figure S18.

Fig S18. Population vector decoding.

(A-C) Left, sample time series of decoded position (dots) and actual position (cyan lines) in the A environment across days 1-4 decoded from day 5. Right, sample decoded vs actual positions across days 1-4 decoded by day 5 reported as probability matrices. (A) Control. (B) $\Delta\text{LEC}_{\text{GLU}}$. (C)

$\Delta\text{LEC}_{\text{GABA}}$. **(D)** A-A decoding error across days relative to day 5 (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.062$, treatment x days, $p = 0.702$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.985$). **(E)** Same as (D) but with A' sessions (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.197$, treatment x days, $p = 0.782$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.377$). **(F)** Same as (D) but with B sessions (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p = 0.001$, days, $p = 0.129$, treatment x days, $p = 0.014$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p = 0.022$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.401$). **(G)** Same as (D) but averaged across sessions (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.116$, treatment x days, $p = 0.314$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.509$). **(H)** A-A decoding error across days between pairs of days (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.694$, treatment x days, $p = 0.747$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.403$). **(I)** Same as (H) but with A' sessions (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.787$, treatment x days, $p = 0.909$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.063$). **(J)** Same as (H) but with B sessions (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.178$, treatment x days, $p = 0.246$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.914$). **(K)** Same as (H) but averaged across sessions (control, $n = 6$, $\Delta\text{LEC}_{\text{GLU}}$, $n = 9$, $\Delta\text{LEC}_{\text{GABA}}$, $n = 8$, two-way ANOVA, treatment, $p < 0.001$, days, $p = 0.551$, treatment x days, $p = 0.788$, Tukey *post hoc* test, control vs $\Delta\text{LEC}_{\text{GLU}}$, $p < 0.001$, control vs $\Delta\text{LEC}_{\text{GABA}}$, $p < 0.001$, $\Delta\text{LEC}_{\text{GLU}}$ vs $\Delta\text{LEC}_{\text{GABA}}$, $p = 0.364$). Error bars represent SEM.

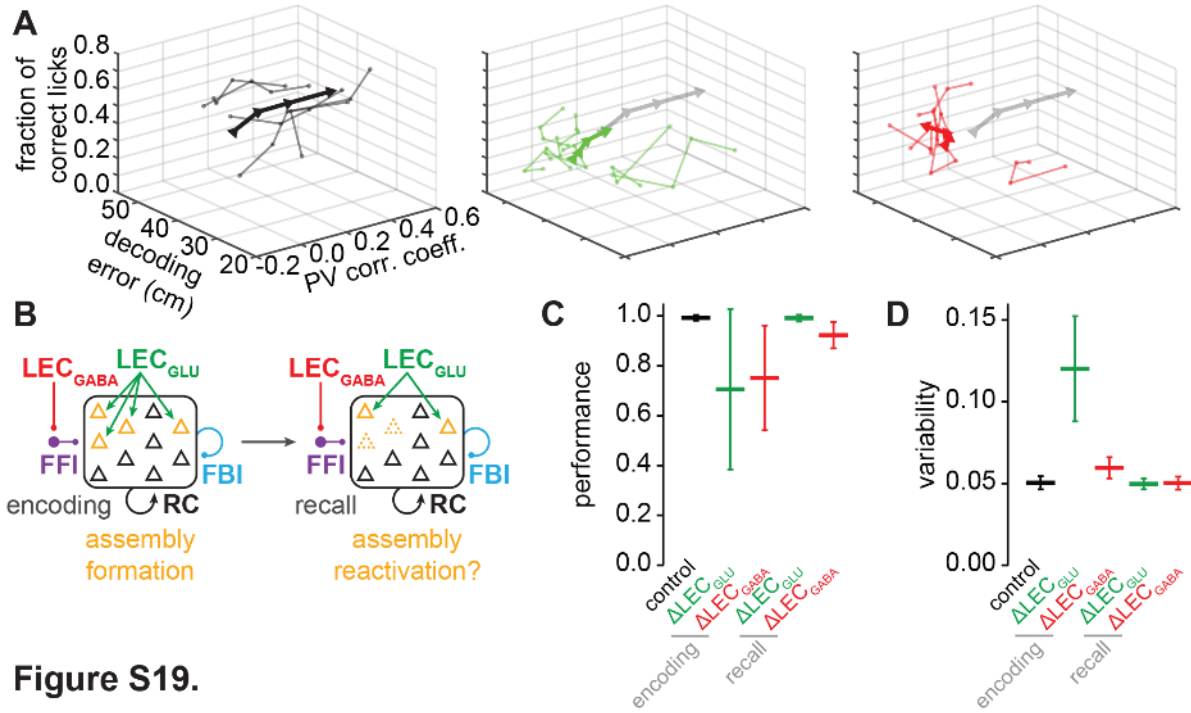


Figure S19.

Fig S19. Modelling LEC_{GLU} and LEC_{GABA} effects on encoding and recall of CA3 ensembles.

(A) Behavioral performance (fraction of correct licks) as a function of neural ensemble stability (PV correlation coefficient) and spatial information (decoding error) in control (black, left), ΔLEC_{GLU} (green, middle; control average shown in gray for reference), and ΔLEC_{GABA} (red, right; control average shown in gray for reference) conditions. (B) Assembly formation and reactivation is modelled in a CA3 neuronal network with recurrent excitation (RC), and lateral feedback inhibition (FBI). LEC_{GLU} provides excitation to a subset of neurons, and LEC_{GABA} drives disinhibition by reducing feedforward inhibition (FFI). (C) Reactivation performance of the modelled neural assembly. (D) Variability of activity in the modelled neuronal network across encoding and recall times. Error bars represent standard deviation.

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